Strategic Action Plan for a Sustainable Positive Energy District (PED)

Campus UNIPD, Padova

Feb 2026

Aliakbar Mokaram Raftari

Tanel Kivipõld

Yaroslav Tochinsky

Andrii Shutenko

Table of contents

1. Technical Strategy Development 4
1.1. Identification of Key Retrofit Measures 4
1.1.1. Objective 4
1.1.2. Strategic Approach: The "Two-Cluster" Methodology 4
1.1.3. Baseline Energy Analysis 4
1.1.4. Intervention Measures & Simulation 4
1.2. Key Technical Assumptions and Considerations 5
1.2.1. Projected Results 5
1.3. Integration of Renewable Energy Sources (Solar PV) 6
1.3.1. Objective 6
1.3.2. Methodology and Assumptions 6
1.3.3. Governing Equations and Parameters 6
1.3.4. Results and District Solar Potential 7
1.4. Annual and/or monthly energy balance 7
1.4.1. Objective 7
1.4.2. Methodology and Assumptions 7
1.4.3. Governing Equations and Parameters 8
1.4.4. Results and District Energy Profile 9
1.5. Definition of Relevant Performance Indicators for PED Monitoring 9
1.5.1. Objective 9
1.5.2. Section A: Component-Level Performance Indicators 9
1.5.3. Section B: District-Level PED Indicators 10
2. UNIPD – Social & Political Feasibility Analysis (5 Chapters) 11
2.1. Stakeholder Ecosystem & Influence Structures 11
2.1.1. Key actors (stakeholder mapping): 11
2.1.2. Key actors: Influence–Interest Interpretation: 12
2.2. Behavioral Patterns & Social Acceptance Dynamics 12
2.3. Equity Considerations 13
2.4. Engagement & Communication Strategy 13
2.4.1. Core Engagement Components 13
2.4.2. Engagement tools for different groups 13
2.4.3. Implementation horizon 14
2.5. Political & Institutional Constraints 14
2.5.1. Heritage Preservation Constraints 15
2.5.2. City of Padova Governance 15
2.5.3. Institutional Fragmentation 15
2.5.4. Financial & Procedural Barriers 15

3. Institutional (policy and regulatory gaps/barriers, governance structure, planning documents) 15
3.1. Governance Structure of UNIPD 15
3.1.1. Institutional Framework 15
3.1.2. Decentralized Operational Structure 16
3.1.3. Administrative Regulations 16
3.2. Institutional Planning Documents Related to Energy & Sustainability 16
3.2.1. Energy Plan 16
3.2.2. Performance Monitoring and Renewable Energy 16
3.2.3. Sustainable Building Initiatives 16
3.3. Policy & Regulatory Gaps and Barriers 16
3.3.1. Historic Building Constraints 16
3.3.2. Fragmented Energy Systems 16
3.3.3. Limited Regulatory Autonomy Over Energy Infrastructure 17
3.3.4. Lack of Formal PED Integration in Core Strategy 17
3.4. Alignment with External Planning Instruments 17
3.4.1. Climate City Contracts and Padova Neutrality Goals 17
3.4.2. EU and National Policies 17
Bibliography 18

Executive Summary

This report outlines a comprehensive, phased strategy to transform the University of Padua (UNIPD) campus into a Positive Energy District (PED). Given the campus's profound heritage constraints and massive user base, a purely technical approach is insufficient. Therefore, this action plan logically integrates four critical dimensions: a targeted engineering retrofit plan (Modern vs. Historic clusters), a robust social and behavioral engagement framework, a strategy to navigate municipal and legal governance barriers, and a €134.2 million financial model utilizing Conto Termico 3.0 and ESCOs. Ultimately, this multipronged approach reduces baseline energy demand, maximizes localized solar generation, and leverages Renewable Energy Communities to achieve true energy positivity.

1. Technical Strategy Development

1.1. Identification of Key Retrofit Measures

1.1.1. Objective

The primary objective of this phase is to systematically evaluate the baseline energy performance of the UNIPD district and identify targeted building retrofit measures to minimize the overall thermal energy demand. Because aggressively reducing energy consumption is the foundational prerequisite for achieving a Positive Energy District (PED), this analysis establishes a realistic, reduced "Target Line" for district energy use prior to the integration of new renewable generation systems.

1.1.2. Strategic Approach: The "Two-Cluster" Methodology

To develop a technically grounded and financially viable retrofit strategy, we analyzed the 26 buildings of the UNIPD district. Given the heterogeneous nature of the building stock—ranging from medieval masonry to modern concrete structures—a "one-size-fits-all" solution is impossible due to strict heritage protection laws.

Therefore, we adopted a Clustering Strategy, dividing the district into two distinct intervention categories:

1. The Modern Cluster (1970s–1990s): Large concrete facilities (e.g., Faculty 5, Faculty 4) with high energy consumption but no heritage constraints. These are targeted for Deep Renovation.
2. The Historic Cluster (Pre-1945): Protected masonry buildings (e.g., Administrative 1, Faculty 1) where external alterations are prohibited. These are targeted for Conservative System Upgrades.

1.1.3. Baseline Energy Analysis

The district's total baseline energy footprint consists of both electrical and thermal loads. Electricity—primarily used for lighting, appliances, and cooling—is supplied by the national grid, totaling 17,736.11 MWh/year across the 26 buildings. The current heating system relies heavily on natural gas. To align thermal energy data with electrical consumption for the total Energy Balance, we first converted the volumetric gas data (Smc) provided in the district profile into energy units (MWh) using the Lower Heating Value (LHV) [1].

$$E_{MWh} = V_{Smc} \times 0.0107_{\left(\frac{MWh}{Smc}\right)}$$

E_{MWH}: The calculated annual energy consumption expressed (MWh)
V_{Smc}: The volume of natural gas consumed annually in Standard Cubic Meters (from district data)
The value 0.0107: The conversion factor derived from the standard Lower Heating Value (LHV) of natural gas. It assumes approximately 10.7 kWh of energy per cubic meter of gas.

1.1.4. Intervention Measures & Simulation

1.1.4.1. A. Modern Cluster: Deep Renovation (Technical & Financial Driver)

The "Modern Giants" (Faculty 5, 4, 6, 7, 8, 12, 13) represent the district's largest energy consumers. For example, Faculty 5 alone consumes ~150,000 Smc of gas annually. Since these buildings are not protected by heritage laws, they offer the highest Return on Investment (ROI) for deep retrofitting.

We modeled the renovation potential using the TABULA WebTool [2], selecting the specific building archetype that matches Faculty 5's construction era and material:

- **Archetype Selected:** IT.MidClim.AB.06.Gen (Apartment Block, 1976–1990).
- **Intervention Level:** "Advanced Refurbishment" (insulation of envelope, window replacement, and thermal bridge correction).

Simulation Results: The TABULA simulation indicates a reduction in primary energy need for heating from 69.5 kWh/m²a (Existing State) to 43.3 kWh/m²a (Refurbished).

$$Saving\ (\%) = \frac{(EP_{existing} - EP_{refurbished})}{EP_{existing}} \times 100\ \approx\ 38\%$$

Saving (%): The percentage of energy reduction achieved through the retrofit intervention,

$EP_{existing}$: Energy Performance (Existing State). The primary energy need for heating of the building in its current condition [kWh/m²a]

$EP_{refurbished}$: Energy Performance (Refurbished State). The predicted primary energy need for heating after applying the "Advanced Refurbishment" measures [kWh/m²a].

Note: We applied this 38% reduction factor to all buildings in the Modern Cluster.

1.1.4.2. B. Historic Cluster: Conservative Systems Upgrade (Social & Political Constraint)

For the remaining buildings (Administrative, Classrooms, Faculty 1–3), external insulation is legally prohibited to preserve the architectural identity of the city center. This represents a hard Political and Social constraint.

Instead of envelope renovation, our strategy focuses on technical system upgrades:

- Replacing obsolete gas boilers with high-efficiency electric Heat Pumps (where space permits).
- Installation of Building Automation and Control Systems (BACS) to optimize usage.

Assumption: Based on standard engineering practice for non-invasive upgrades, we applied a conservative 15% reduction factor to the Historic Cluster.

1.2. Key Technical Assumptions and Considerations

To perform these calculations within the available dataset, the following justified estimations were applied:

Archetype Approximation: Because the standard TABULA WebTool primarily models residential building stock, the 'Apartment Block' (AB) archetype was utilized as a functional proxy for the Modern Cluster's university faculty buildings, assuming similar concrete construction methods and envelope degradation for the 1970s–1990s era.

Uniform Retrofit Application: The calculated 38% primary energy reduction for the Modern Cluster and the conservative 15% reduction for the Historic Cluster were applied uniformly to all buildings within their respective categories, establishing a high-level district average rather than a building-by-building dynamic simulation.

Thermal Focus: The baseline calculation and subsequent retrofit savings focus strictly on the natural gas heating demand. The natural gas Lower Heating Value (LHV) was assumed constant at 10.7 kWh/Smc, and the baseline electrical demand is assumed to remain constant during this retrofit phase.

Absolute Heritage Constraints: For the Historic Cluster, it was assumed that local architectural preservation laws strictly prohibit any external envelope modifications, restricting interventions entirely to internal mechanical system upgrades (Heat Pumps and BACS).

1.2.1. Projected Results

By applying these targeted saving factors to the baseline consumption, we calculated the new projected energy demand for the district.

$$D_{new}\ = D_{baseline}\ \times (1 - Saving)$$

D_{new}: The projected annual energy demand of the building after renovation measures are applied [MWh], $D_{baseline}$: The current annual energy demand of the building before any intervention [MWh].

Final Outcome: Implementation of this two-tier strategy reduces the district gas demand significantly to 6,584.16 (MWh/year). By combining this newly reduced heating demand with the constant baseline electrical load, which is 17736.11 (MWh/year), the Total New District Energy Demand is established at 24,320.27 (MWh/year) (down from a baseline of 26,893.52 (MWh/year)). This reduced total energy demand serves as the definitive "Target Line" for our Renewable Energy integration in the next phase.

Cluster Type	Targeted Intervention	Expected Energy Savings
Modern Cluster (1970s–1990s)	**Deep Renovation (Insulation & Windows)**	**38%**
Historic Cluster (Pre-1945)	**Conservative Upgrades (Heat Pumps & BACS)**	**15%**
Baseline Total Energy (Gas + Elec)	**26,893.52 (MWh/year)**	
New Total Energy Demand	**24,320.27 (MWh/year)**	

Table 1: District Clustering Strategy and Projected Energy Savings, and Total Baseline and New Energy Demand

1.3. Integration of Renewable Energy Sources (Solar PV)

1.3.1. Objective

The objective of this phase is to evaluate the potential for renewable energy generation within the university district to offset overall energy demand. Given the urban constraints, rooftop Solar Photovoltaic (PV) systems were selected as the primary renewable technology.

1.3.2. Methodology and Assumptions

To accurately estimate the solar potential across the district, several engineering assumptions and constraints were applied:

- **Heritage Constraints (Cluster Selection):** Due to strict historical preservation laws in the center of Padua, installing solar panels on the Historic Cluster buildings is not permissible. Therefore, the solar PV analysis is restricted exclusively to the 7 buildings within the Modern Cluster (Faculty 4, 5, 6, 7, 8, 12, and 13).
- **Roof Area Estimation:** The provided dataset supplies only the Total Heated Floor Area for each building. Because the buildings are anonymized, exact physical roof measurements via GIS mapping tools could not be obtained. To estimate the roof footprint, it is assumed that the Modern Cluster buildings have an average height of 4 stories.
- **Usable PV Area:** A standard flat commercial roof contains HVAC equipment, elevator shafts, and maintenance walkways. To account for this, a standard utilization factor of 50% (0.5) is applied to the estimated roof area to determine the actual space available for solar panel installation.
- **System Sizing and Yield:** It is assumed that modern commercial solar panels require approximately 5 m² of space to generate 1 kilowatt-peak (kWp) of power [3]. The specific annual solar yield for Padua was determined using the European Commission's PVGIS tool [4].

1.3.3. Governing Equations and Parameters

The calculation for the annual solar energy production follows a four-step sequence for each building.

Step 1: Estimated Roof Area

$$A_{roof} = \frac{A_{total}}{N_{floors}}$$

A_{roof} = Estimated Roof Area (m²)
A_{total} = Total Heated Floor Area from the UNIPD dataset (m²)
N_{floors} = Assumed number of building stories (4)

Step 2: Usable PV Area

$$A_{usable} = A_{roof} \times U_f$$

A_{usable} = Usable area available for PV installation (m²)
U_f = PV Utilization Factor accounting for roof equipment and shading (0.5)

Step 3: Solar System Size

$$P_{sys} = \frac{A_{usable}}{S_{req}}$$

P_{sys} = Maximum installed peak power of the solar system (kWp)
S_{req} = Space required per installed kWp (5 m²/kWp)

Step 4: Annual Solar Production

$$E_{solar} = \frac{(P_{sys} \times Y_{solar})}{1000}$$

E_{solar}= Total annual solar energy production (MWh/year)
P_{sys} = System size (kWp)
Y_{solar} = Specific Solar Yield for Padua, extracted from PVGIS (1367.47 kWh/kWp/year)
1000 = Conversion factor from kWh to MWh

1.3.4. Results and District Solar Potential

Applying this methodology to the Modern Cluster, the total usable rooftop area for solar generation is 14,415 m². This allows for an aggregate installed capacity of 2,883 kWp. Based on the local solar yield in Padua (1367.47 kWh/kWp/year), the rooftops of the Modern Cluster can generate a combined total of 3942.42 MWh/year of clean, renewable electricity. This generated electricity will be directly utilized to offset the district's grid dependency.

System Parameter	Calculated Value
Total Usable Roof Area (m²)	14,415
Installed Peak Capacity (kWp)	2,883
Specific Solar Yield (Padua)(kWh/kWp/year)	1,367.47
Total Annual Solar Generation (MWh/year)	3,942.42

Table 2: Solar PV System Sizing and Annual Generation Potential

1.4. Annual and/or monthly energy balance

1.4.1. Objective

The objective of this phase is to evaluate the energy self-sufficiency of the district by calculating the "Balance" between the Total District Energy Demand (reduced heating plus baseline electricity) and the newly integrated renewable solar production. This balance is analyzed both on an annual basis and distributed monthly to reveal seasonal energy dependencies.

1.4.2. Methodology and Assumptions

Because district-level energy consumption and generation fluctuate drastically across the year, an annual aggregate is insufficient for grid planning. A proportional weighting methodology was applied to distribute the annual totals into a monthly profile:

- **Heating Demand Distribution:** The exact monthly heating consumption was dynamically calculated using the standard Heating Degree Days (HDD) for Padua, Italy [5]. Using the normative weather data

provided in the project framework, the total heating load was heavily weighted toward the winter months (peaking in January and December) and strictly set to zero during the summer months (June through August) when heating systems are offline.

- **Electricity Demand Distribution:** Due to the lack of month-by-month electrical metering data, the annual electrical demand (17,736.11 MWh) was distributed evenly across the 12 months (1,478.01 MWh/month), acting as a constant district base-load for continuous services (e.g., servers, security, and lab equipment).
- **Solar Production Distribution:** Solar generation is highly dependent on seasonal irradiance. The annual PV generation was distributed across the 12 months using the exact monthly yield profile (E_m) extracted from the European Commission's PVGIS tool for the specific coordinates of the Padua district.
- **The "Gap" Visualization:** The final monthly balance was visualized using a clustered column chart to clearly contrast energy consumption against PV energy generation, matching standard urban building energy modeling (UBEM) reporting formats.

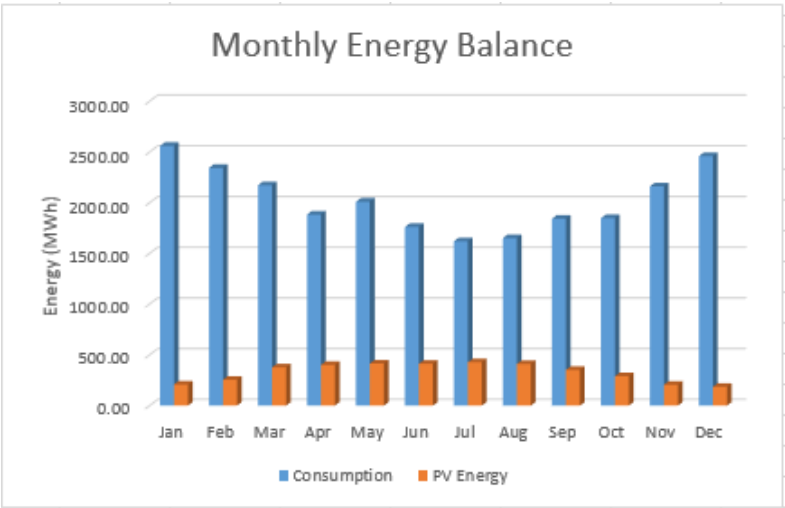

Figure 1: Monthly District Energy Balance, Total Energy Demand (consumption) vs. Solar PV Generation

1.4.3. Governing Equations and Parameters

The district's energy balance is defined by three primary equations.

Step 1: Annual Energy Balance

$$E_{Balance} = E_{solar} - E_{demand}$$

$E_{balance}$ = Annual Energy Deficit requiring grid import (-20,377.85 MWh/year)
E_{demand}= Total District Energy Demand [Heating + Electricity] (24,320.27 MWh/year)
E_{solar} = Total District PV Solar Production (3942.42 MWh/year)

Step 2: Monthly Heating Demand (Proportional Weighting)

$$D_{month} = E_{demand} \times \left(\frac{HDD_{month}}{HDD_{total}}\right)$$

D_{month} = Calculated heating demand for a specific month (MWh)
HDD_{month} = Standard Heating Degree Days for the specific month in Padua
HDD_{total} = Total annual Heating Degree Days for the region (3409)

Step 3: Monthly Solar Production (Proportional Weighting)

$$S_{month} = E_{solar} \times (\frac{Y_{month}}{Y_{total}})$$

S_{month} = Calculated solar energy production for a specific month (MWh)
Y_{month} = Monthly specific solar yield for Padua extracted from PVGIS (kWh)
Y_{total} = Total annual specific solar yield extracted from PVGIS (1367.47 kWh)

1.4.4. Results and District Energy Profile

The annual energy balance reveals that the district produces 3,942.42 MWh/year while consuming a total of 24,320.27 MWh/year, resulting in a net annual deficit (Gap) of 20,377.85 MWh. Furthermore, the monthly energy balance chart reveals a continuous energy dependency with a severe seasonal peak. Even during the summer months—when heating demand drops to zero and solar production reaches its maximum—the constant electrical base-load still exceeds PV generation. In deep winter (December and January), the gap widens significantly as solar production drops to its lowest point exactly when heating demand spikes to its maximum. This proves that while rooftop PV heavily offsets annual carbon emissions, the dense historic district fundamentally relies on the national grid year-round.

1.5. Definition of Relevant Performance Indicators for PED Monitoring

1.5.1. Objective

To successfully monitor the Padua district's transition toward Positive Energy District (PED) status, a rigorous framework of quantifiable Key Performance Indicators (KPIs) must be established. This monitoring framework is divided into two tiers: Component-Level indicators (measuring the performance of individual retrofits and systems) and District-Level indicators (measuring the macro-performance of the neighborhood as a single entity). Where applicable, the district's current baseline performance has been calculated using the master energy balance.

1.5.2. Section A: Component-Level Performance Indicators

Before district-wide positivity can be achieved, the individual physical systems must perform optimally. The following indicators track the specific interventions applied to the buildings and roofs.

1.5.2.1. Energy Savings Percentage (Building Level)

Measures the reduction in specific heating demand due to the implementation of passive architectural retrofits (e.g., insulation, window replacements) compared to the historical baseline.

$$Savings = \frac{(E_{baseline} - E_{retrofit})}{E_{baseline}} \times 100$$

S_{avings} = Energy savings percentage (\%)
$E_{baseline}$ = Original heating demand before interventions (MWh/year)
$E_{retrofit}$= New heating demand after passive architectural improvements (MWh/year)

Current Status: A highly successful 38% reduction for modern buildings and a 15% reduction for historic buildings, successfully capping total district heating demand at 6584.16 MWh/year.

1.5.2.2. Specific PV Yield (System Level)

Evaluates the generation efficiency of the installed solar photovoltaic system in the specific geographic location, acting as a health monitor for the solar hardware over time.

$$Y_{specific} = \frac{E_{solar_annual}}{P_{installed_capacity}}$$

$Y_{specific}$ = Specific PV yield (kWh/kWp/year)
E_{solar_annual} = Total annual energy generated by the solar system (kWh/year)
$P_{installed_capacity}$ = Total installed peak power of the PV panels (kWp)

Current Status: 1367.47 kWh/kWp/year, establishing a strong baseline for renewable generation in Padua based on regional climate data.

1.5.3. Section B: District-Level PED Indicators

These five core indicators define the ultimate success of the PED strategy, monitoring the aggregated interaction between the district's energy demands and local generation.

1.5.3.1. Annual Energy Balance

The absolute difference between the total energy consumed by the district's buildings and the total renewable energy generated on-site over one year. A true PED requires this value to be greater than zero.

$$E_{balance} = E_{generation} - E_{demand}$$

$E_{balance}$ = Annual energy balance or deficit (MWh/year)
$E_{generation}$ = Total renewable energy produced locally by the district (MWh/year)
E_{demand} = Total energy consumed by the district (MWh/year)
Current Status: 3942.42 MWh/year (Solar PV) - 24,320.27 MWh/year (Total Energy Demand) = -20,377.85 MWh/year.

Analysis: The district currently operates at a significant energy deficit. Further interventions (e.g., heat pumps, deep facade retrofits, and extensive renewable capacity expansion) are required to fully close this gap.

1.5.3.2. Renewable Energy Share (RES) / Self-Sufficiency

The percentage of the district's total energy demand that can theoretically be satisfied by its own local renewable energy generation.

$$RES = \left(\frac{E_{generation}}{E_{demand}}\right) \times 100$$

RES = Renewable Energy Share or Self-Sufficiency (%)
$E_{generation}$ = Total renewable energy produced locally (MWh/year)
E_{demand} = Total energy consumed by the district (MWh/year)
Current Status: (3942.42 / 24,320.27) × 100 = 16.2%.

Analysis: The proposed rooftop PV integration covers approximately 16.2% of the district's total energy needs on an annual basis.

1.5.3.3. Self-Consumption Rate (SCR)

The proportion of the locally generated renewable energy that is actually consumed directly by the district itself, rather than being exported to the wider national grid.

$$SCR = \left(\frac{E_{PV_used_locally}}{E_{PV_generated}}\right) \times 100$$

SCR = Self-Consumption Rate (%)
$E_{PV_used_locally}$ = Amount of solar energy consumed directly by the district's buildings (MWh/year)
$E_{PV_generated}$ = Total solar energy generated by the district's roofs (MWh/year)

Current Status: Exact calculation requires hourly dynamic simulations. However, the monthly balance indicates near 100% SCR year-round. Because the constant electrical base-load significantly exceeds the peak summer solar production, virtually all generated PV energy is consumed directly by the district's buildings without needing to export to the grid.

1.5.3.4. CO_2 Emission Reduction

The total volume of greenhouse gases avoided due to the implementation of the district retrofit and renewable generation strategies.

$$\Delta CO_2 = (E_{saved} \times EF_{gas}) + (E_{solar} \times EF_{grid})$$

ΔCO_2 = Total reduction in carbon dioxide emissions ($kgCO_2$/year)
E_{saved} = Energy saved from heating demand reduction (kWh/year)
EF_{gas} = Emission factor of the displaced heating fuel, such as natural gas ($kgCO_2$/kWh) [6]
E_{solar} = Renewable energy generated by the PV system (kWh/year)
EF_{grid} = Emission factor of the national electricity grid ($kgCO_2$/kWh) [6]

Monitoring Strategy: Progress is continuously monitored by tracking the reduction of natural gas combustion on-site, compounded by the clean energy injected by the 2.88 MWp PV system.

1.5.3.5. Resilience Index

A measure of the district's ability to maintain critical operational loads during external grid shocks, winter storms, or power outages.

Monitoring Strategy: Evaluated by calculating "hours of autonomy." Currently, the district's winter resilience is low due to a reliance on imported grid energy to cover the 20,377.85 MWh deficit. Future steps to improve this indicator must include thermal or electrical storage systems and demand-side management.

Indicator Name	Current Status
Energy Savings Percentage	38% (Modern) / 15% (Historic)
Specific PV Yield (kWh/kWp/year)	1,367.47
Annual Energy Balance (MWh/year)	-20,377.85 (Deficit)
Renewable Energy Share (RES)	16.2%
Self-Consumption Rate (SCR)	Near 100% (Year-Round)

Table 3: Positive Energy District (PED) Key Performance Indicators Summary

2. UNIPD – Social & Political Feasibility Analysis (5 Chapters)

2.1. Stakeholder Ecosystem & Influence Structures

UNIPD is a unique city campus university where the entire city functions as the campus. As a result, Positive Energy District (PED) transformation is not only a technical challenge but a social and institutional process that affects thousands of users. The university's spatial footprint spans more than 791,000 m² and includes educational buildings, laboratories, administrative facilities, museums and heritage complexes.

In order to effectively plan and implement PED-related strategies, it is essential to systematically map the key actors who shape, enable, or are directly affected by the transition. Stakeholder mapping helps clarify the distribution of power, interests, and responsibilities across the ecosystem, allowing project designers to identify where strong collaboration is required, where engagement efforts must be intensified, and where barriers may arise. By positioning actors along an influence–interest axis, it becomes possible to prioritise decision-making partners, anticipate conflicts or misalignments, and tailor participation strategies that reflect each group's capacity to affect PED outcomes.

2.1.1. Key actors (stakeholder mapping):

Students (ca 75 000) [11]**,** the largest user group with high impact on energy outcomes via behaviour, yet low formal decision power.
Academic Staff (~2 700) [11] – strong impact on HVAC and laboratory loads; high expectations of autonomy. a relatively large user group capable of exerting direct influence, acting as frontrunners, and setting an example for others.
Administrative & Technical Staff (~2 800) [11] – actual operational control.
City of Padova – responsible of energy, transport and urban planning policies.
Heritage Authorities – critical decisionmakers for retrofits, PV and facade solutions.
Local Grid operators – Hera Group, a major distribution system operator responsible for electricity, water, and district heating networks in Padova [9] —play a critical role in shaping the development of the local distribution

infrastructure and determining the technical and regulatory conditions for feeding surplus electricity back into the grid

Accommodation providers – including student dormitories as well as institutional partners such as ESU (Housing), which operates 21 residences with approximately 1,650 beds—constitute a significant stakeholder group due to their substantial influence on residential energy-efficiency performance.[7]

Public Visitors – affect the use of buildings, limited influence and minimal engagement incentives.

2.1.2. Key actors: Influence–Interest Interpretation:

High influence / High interest: City, Heritage Authority, Academic Staff, Administrative & Technical Staff → key decisionmakers.

High interest / Medium influence: Students → strong engagement programmes are required.

Medium influence / Medium interest: Accommodation providers, grid operators →necessary to cooperate

Low influence / Medium interest: public visitors → small and short-term impact

2.2. Behavioral Patterns & Social Acceptance Dynamics

UNIPD's energy profile is heavily shaped by user behaviour because many buildings are historic and rely on natural ventilation, manual controls and open door/window practices. Heritage buildings (Palazzo Bo, Maldura, Liviano, etc.) face technical constraints, which means behavioural patterns play a larger role than in modern campuses.

1) Students

- Long study days → high ventilation and lighting loads.
- Irregular window opening (significant winter heat loss).
- High plug load (laptops, chargers).
- High cooling expectations in summer (peak days up to 30–40 °C).

2) Academic & research staff

- Laboratories operate long hours, often 24/7 (e.g., Pharmaceutical Sciences Institute).
- High ventilation demand requires continuous air exchange.
- Potential resistance if measures threaten experimental stability or research quality.

3) Administrative staff

- Stable working schedules.
- Potential to standardise energy aware routines.
- Can be key actors in building level optimisation.

It is important not to underestimate the impact that building occupants have on overall energy consumption. For example, the article "Occupant Behavior Impact on Building Sustainability Performance: A Literature Review" (Sustainability, 2023) reports that:

- Adjusting thermostat settings by ±2°C can affect heating and cooling energy use by 10–30%. (Habtamu Tkubet Ebuy, 2023)
- Opening windows can increase heating demand by up to 50% in cold climates, and cooling demand by 10–20% in warm regions. (Habtamu Tkubet Ebuy, 2023)
- Switching off lights and appliances can reduce electricity consumption by 10–25%. (Habtamu Tkubet Ebuy, 2023)

The authors emphasize that the discrepancy between the designed and actual energy use of buildings — the so-called performance gap — is attributable to occupant behaviour by at least 30–50%. (Habtamu Tkubet Ebuy, 2023)

It is also important to identify potential high-risk areas related to social acceptance, for example:

- Limiting window opening – may cause additional discomfort if ventilation is not sufficient
- Lowering winter setpoints – may reduce work productivity and may cause health issues
- PV and ventilation solutions that alter protected facades – the aim to preserve the architectural values of buildings
- Changing laboratory operating regimes - may not be feasible given the constraints of laboratory experiments

Therefore, it is essential to identify which actions or interventions influence occupant comfort and well-being, and to prioritise measures that do not adversely affect users while still enabling meaningful progress toward the intended objectives.

2.3. Equity Considerations

UNIPD's user environment is socially diverse: 70,000+ students (including international students), learners with different household incomes, staff with diverse needs, and a building stock spanning medieval to modern Nearly Zero-Energy Building (NZEB) level facilities with very different comfort profiles.

Key equity topics

- **Thermal Comfort Equity**: historic stock (e.g., Palazzo Bo, Maldura) is cold in winter and hot in summer, while newer facilities (e.g., Turazza Residence) are NZEB level → uneven indoor quality.
- **Cost Equity**: students in residences face rising energy costs; some savings measures may feel "fair" yet reduce comfort.
- **Accessibility**: heritage buildings limit barrier free retrofits; interventions must balance energy and access.
- **Digital Equity**: dashboards/Apps for engagement must work for all users, including those with limited digital proficiency.
- **Participation Equity**: ensure an "equal voice" beyond senior faculty—for students, technical staff and residence users.

2.4. Engagement & Communication Strategy

PED transformation at UNIPD must progress with all user groups. The entire university (educational buildings, heritage complexes, labs, residences, sports facilities) is socially and functionally intertwined.

2.4.1. Core Engagement Components

Effective stakeholder engagement within a PED context relies on a set of foundational components that guide how information is shared, how collaboration is structured, and how trust is built over time. These components help ensure that diverse actor groups not only understand the rationale behind the transition but are also meaningfully involved in shaping solutions, providing feedback, and adapting practices. By structuring engagement around clear principles—ranging from awareness-building to evidence-based communication—the project can establish consistent, transparent, and inclusive processes that support long-term behavioural and operational change.

- **Awareness**: explain the "why" (climate, economics, heritage constraints).
- **Cocreation**: involve students and faculty early.
- **Feedback loops**: building level dashboards and reporting channels.
- **Localization**: tailor strategies to building typologies (heritage vs modern).
- **Evidence based**: share data to build trust.

2.4.2. Engagement tools for different groups

To engage different groups as effectively as possible, it is necessary to develop tailored awareness-raising programmes that best fit each group, ensuring a successful engagement outcome.

Students

- Green Ambassadors programme - A student-led sustainability team trained to promote energy-aware behaviour across campus. This is one of the most effective engagement formats in universities because behaviour change spreads peer-to-peer, not top-down.
- Residence energy gamification - by using competition, feedback, and rewards to shape habits in a fun and motivating way.
- Short "Energy 101" bootcamps - A simple but highly effective introduction to energy literacy for students—especially international students and first-years living in residences.

Academic Staff

• Fume hood setback and "close the sash" policies - To reduce excessive ventilation and energy losses in research laboratories by encouraging safer, more efficient use of chemical fume hoods
• Lab operational guidelines and codesign workshops - To collaboratively develop clear, practical operational guidelines for laboratories that maintain safety and research quality while identifying energy-intensive practices that can be optimised.
• Structured "Comfort vs Safety vs Energy" dialogues - To facilitate ongoing, transparent discussions that help align three often conflicting priorities in university buildings:

- occupant comfort
- regulatory safety requirements
- energy-efficiency expectations

These dialogues support informed decisions, reduce conflicts, and build trust between academic staff and facility managers.

Administration

• Training for building energy diagnostics - To equip administrative and technical staff with the skills needed to identify inefficiencies, detect abnormal energy patterns, and respond quickly to operational problems that increase energy consumption.
• Standard operating procedures (SOP) for seasonal changeovers and setpoints - To harmonise building operation across the entire university by defining clear procedures for switching between heating and cooling seasons, and for managing temperature setpoints in a consistent, efficient manner.

City & Grid operators & Heritage Authorities

• Joint coordination forums - To create a stable governance mechanism that allows the University of Padua, the City of Padua, regional representatives, and heritage protection authorities to coordinate energy-related decisions, share data, align planning horizons, and avoid regulatory conflict.
• Catalogue of heritage compatible retrofit solutions - To develop a jointly endorsed technical reference catalogue listing retrofit measures that comply with heritage restrictions while improving energy performance. This catalogue streamlines renovation processes, reduces negotiation time, and provides design teams with clear guidance.

2.4.3. Implementation horizon

For the PED transformation to be effective, it is essential to establish a clearly structured and realistic action plan, ensure systematic monitoring of its implementation, and accelerate any components that show signs of delay. Rather than setting overly ambitious or unattainable deadlines, the planning process should include a careful assessment of feasible timeframes, the definition of a coherent long-term roadmap, and the subdivision of this roadmap into manageable and interdependent sub-projects. The EU Joint Research Centre's report 'Opportunities and challenges for EU energy policy 2025–2035: a systemic perspective' (2025) likewise emphasises that transition projects must be grounded in realistic planning assumptions, as overly ambitious or insufficiently structured approaches are prone to failure.[10]

Therefore, propose structuring the approach into three broad categories.

• **Short-term (0–1 years)**: awareness, baseline audits, stakeholder alignment.
• **Medium-term (1–3 years)**: retrofitted pilots, cocreated interventions.
• **Long-term (3–10 years)**: deep renovation, district scale integration.

2.5. Political & Institutional Constraints

UNIPD's sociopolitical environment is multilayered and complex.

2.5.1. Heritage Preservation Constraints

Some UNIPD buildings are several centuries old, thus PV, facade insulation or window replacement is often either:

(a) fully restricted.
(b) allowed only in nonvisible zones.

(c) permitted subject to special committee approvals (e.g., Palazzo Bo, Palazzo Maldura, Palazzo del Capitanio).

2.5.2. City of Padova Governance

The city controls: energy management, district heating regimes, public realm/permits, mobility. Therefore, the PED pathway must align with municipal energy and climate plans.

2.5.3. Institutional Fragmentation

UNIPD spaces belong to the University, ESU, private landlords (leased space), and multiple heritage bodies. (University of Padua. , 2026). The "multi owner problem" slows renovation. Therefore, it is necessary to find solutions for involving different owners and to provide funding schemes or motivation mechanisms that encourage their participation in achieving the PED objectives.

2.5.4. Financial & Procedural Barriers

2.5.4.1. Multiyear budgeting cycles.

Long planning and budgeting horizons—often spanning several years—make it difficult to rapidly mobilize financial resources for PED-related interventions. Universities and public institutions typically allocate funds well in advance, which limits flexibility and slows down the implementation of innovative or time-sensitive projects.

2.5.4.2. National bureaucracy for energy grants.

Accessing national or EU-level funding for energy efficiency, renewable energy, and sustainability upgrades is frequently constrained by complex administrative procedures. Lengthy application processes, strict eligibility requirements, and extensive reporting obligations can delay project timelines and discourage smaller stakeholders or private building owners from applying.

2.5.4.3. Multilevel coordination (University + City + Region + State + EU)

PED development requires close cooperation across several layers of governance, each with its own priorities, regulations, and approval processes. Aligning the strategic goals, regulatory frameworks, and funding mechanisms of the university, municipal government, regional authorities, national ministries, and EU institutions introduces procedural complexity and can lead to delays, fragmented decision-making, or duplicated efforts. Institutional (policy and regulatory gaps/barriers, governance structure, planning documents)

3. Institutional (policy and regulatory gaps/barriers, governance structure, planning documents)

3.1. Governance Structure of UNIPD

3.1.1. Institutional Framework

UNIPD operates under the authority of an academic statute that defines its organization, autonomy, and main decision-making bodies. The statute is approved by the Italian government's Ministry of Education and Research and reflects national university law.

Core governance bodies include:

- Academic Senate – academic deliberative body
- Board of Directors – administrative and financial governance
- Rector and Central Administration – top executive functions
  These entities make strategic choices related to sustainability and campus infrastructure.

3.1.2. Decentralized Operational Structure

Beyond the central governance, individual departments and faculties have autonomy in research and teaching priorities, which means sustainability initiatives must be coordinated across diverse units.

3.1.3. Administrative Regulations

UNIPD maintains a wide set of internal regulations that shape governance practices but do *not* specifically mandate energy-transition or climate objectives.

3.2. Institutional Planning Documents Related to Energy & Sustainability

UNIPD has developed strategic alignment mechanisms toward environmental performance and energy planning:

3.2.1. Energy Plan

In response to the Italian Ministerial Decree 383/2022 on energy containment, UNIPD adopted an internal energy plan to reduce energy consumption and greenhouse gas emissions.

This plan includes:

- Short-term measures
- Mid- and long-term energy efficiency upgrades
- Annual monitoring and performance indicators

3.2.2. Performance Monitoring and Renewable Energy

UNIPD tracks energy performance across buildings and has shifted to electricity solely from renewable sources since 2019, cutting CO_2 emissions from electricity consumption.

The institution also participates in the Renewable and Solidarity Energy Communities (RSEC) initiative with the City of Padova to explore local renewable energy communities — a governance/energy policy innovation that bridges university and municipal planning.

3.2.3. Sustainable Building Initiatives

The university's sustainable building program is significant, with ~€420 million allocated for construction and renovation to meet LEED and near-zero energy building standards.

3.3. Policy & Regulatory Gaps and Barriers

UNIPD faces *institutional and external obstacles* that affect its ability to transition toward a Positive Energy District:

3.3.1. Historic Building Constraints

Many of UNIPD's buildings are historic and located in the city centre. These structures often fall under cultural heritage protection laws that restrict facade upgrades, external insulation, solar installations, and structural modifications.

While this heritage value enriches the university, it limits compliance with energy-efficient upgrades and PED infrastructure unless exemptions or special permissions are granted — a clear policy conflict between preservation and energy transformation.

3.3.2. Fragmented Energy Systems

UNIPD's building stock is highly heterogeneous (historic and modern buildings scattered in urban areas). This complicates the application of uniform regulatory standards and energy retrofitting plans typical for compact campuses.

National and EU directives push toward higher energy performance and efficiency metrics — but implementation in a diverse building stock requires custom strategies, which existing policies do not uniformly address.

3.3.3. Limited Regulatory Autonomy Over Energy Infrastructure

Universities can plan internal energy efficiency, but energy supply infrastructure is regulated at municipal/national level. UNIPD cannot independently establish micro-grids or public shared energy generation without navigating:

- Grid regulations
- Local energy planning frameworks
- Permits for renewables

These are external barriers embedded in the larger energy regulatory system.

3.3.4. Lack of Formal PED Integration in Core Strategy

Although UNIPD engages in sustainable energy projects, there is no formal, comprehensive institutional policy that defines:

- A university-wide PED target
- Governance roles for PED implementation
- Coordination mechanisms with the City of Padova's climate plans

This gap means sustainability efforts are often project-driven, not part of formal institutional planning.

3.4. Alignment with External Planning Instruments

UNIPD's energy goals intersect with broader urban and regional strategies:

3.4.1. Climate City Contracts and Padova Neutrality Goals

UNIPD collaborates with the City of Padova's initiatives, such as the Climate City Contract under the EU's Net Zero Cities mission — aimed at achieving climate neutrality by 2030.

These joint documents are crucial for aligning university energy planning with city-wide decarbonization and potentially PED deployment.

3.4.2. EU and National Policies

UNIPD's plans respond to EU climate targets and national energy efficiency requirements for buildings and public institutions.

However, institutional policy frameworks lag behind these broader mandates, meaning there is a *regulatory adoption gap* at the university level.

4. Financial Assessment

4.1 CAPEX Calculation for PED UNIPD Strategy

The calculation of capital expenditures (CAPEX) for the PED UNIPD strategy is based on the differentiation of buildings by the type of required interventions. The Modern cluster, consisting of facilities from the 1970s–1990s, requires "deep renovation," while the Historical cluster focuses on "conservative modernization of systems." This strategy avoids legal conflicts with heritage protection authorities and ensures the maximum technical result for every euro invested.

Modern Cluster:

For modern buildings, such as faculties 4, 5, 6, 7, 8, 12, and 13, capital expenditures cover full insulation of facades and roofs, as well as the replacement of window systems.

- Total area: Approximately 115,320 m².
- Market data: Renovation data for Italy in 2024–2025 indicates an average cost for deep renovation of 800 EUR/m².

Historical Cluster:

For the Historical cluster, where facade work is restricted, CAPEX is concentrated on internal modernization. This includes:

- Replacement of hydraulic networks.

- Installation of low-temperature emitters.
- Local internal insulation (where permitted).
- Estimated cost: 350 EUR/m².

System Upgrades and Automation:

Heat Pumps (HP):

The transition to heat pumps is the most critical step in eliminating natural gas consumption. Across the campus average, considering the necessary hydraulic work and adaptation of existing networks, CAPEX for heat pumps ranges from 80–120 EUR/m² for standard renovations.

Building Automation and Control Systems (BACS): BACS are critical for optimizing consumption and managing the PED.

- Cost: Approximately 5 EUR/m² for educational buildings.
- Payback period: These systems have the shortest ROI—typically less than 3-5 years—due to the immediate reduction of energy waste.

Taking into account the need for digital management system integration and heating system modernization, we adopt a weighted average of 900 EUR/m² for the Modern cluster and 350 EUR/m² for the Historical cluster.

Table 1. CAPEX

Type of Measures	Target Buildings	Specific Cost (EUR/m² or EUR/kW)	Determining Parameter	CAPEX (EUR)
Deep Renovation (Building insulation, automated energy management, heat pump installation)	Modern buildings, Faculty (4-8, 12, 13)	900	115,320 m²	103,788,000.00
Renovation (Energy management, hydraulic network replacement, local internal insulation, heat pump installation)	Historical buildings, Administrative, Classrooms, Faculty 1–3	350	78,716 m²	27,550,600.00
PV System Installation (Solar panels)	Faculty (4-8, 12, 13)	1,000	2,883 kW	2,883,000.00
Total				134,221,600.00

CAPEX – 134 211 600 EUR.

4.2 Impact of the PED Strategy on Operational Expenditures (OPEX)

The impact of the PED (Positive Energy District) strategy on the university's operational expenditures has a twofold nature. On one hand, there is a radical reduction in costs for purchasing energy resources (natural gas and grid electricity). On the other hand, new expenditure items emerge for the technical maintenance of high-tech equipment and digital systems. Accordingly, 1% of the total investment can be allocated; these funds will be used for the servicing of new engineering systems and the decommissioning (dismantling) of old systems.

OPEX – 1 342 116 EUR.

4.3 Financing Mechanisms

One of the most critical aspects of the financial assessment is the identification of external capital sources. For the University of Padua, as a public institution, a wide range of instruments is available that can cover between 40% and 100% of capital expenditures.

4.3.1. Conto Termico 3.0: The Primary Project Driver

The Conto Termico 3.0 program, effective from December 2025, is a key tool for Italian Public Administrations (PA). It allocates €400 million annually specifically for the public sector.

- Support Level: The university can expect coverage of up to 65% of costs for the energy modernization of buildings.
- Special Conditions: For facilities converted into Nearly Zero-Energy Buildings (NZEB), the grant level may reach 100% under certain conditions (e.g., for schools and hospitals, which may include the university's clinical buildings).
- Payment Mechanism: Unlike tax credits, Conto Termico provides direct cash payments to the institution's bank account over a period of 2–5 years.

4.3.2. EIB Programs and ELENA

The University of Padua can utilize technical assistance from the ELENA (European Local ENergy Assistance) program, which covers up to 90% of the costs for developing project documentation, energy audits, and tender preparation. Furthermore, direct lending from the European Investment Bank (EIB) for university infrastructure (as seen with the Politecnico di Torino) provides access to capital with low interest rates and long grace periods.

4.3.3. ESCO Models and PPPs

Utilizing the Energy Performance Contracting (EPC) model allows the university to attract private capital to implement the PED. In this scheme, an ESCO (Energy Service Company) invests its own funds into equipment and works, guaranteeing a specific level of savings.

- Advantage: Capital expenditures do not appear on the university's balance sheet (off-balance sheet), which is crucial given the strict budget constraints of the public sector.
- Incentives: Italian legislation (D.Lgs. 36/2023) allows the use of Public-Private Partnership (PPP) models for deep renovation projects, where the private partner assumes the risks of design, construction, and operation for 15–20 years.

4.4 Investment Logic and Risks

Despite the high potential for savings and the availability of grants, the PED UNIPD strategy faces several financial risks that require careful management.

4.4.1. Regulatory and Procedural Risks

The multi-layered governance structure (University, City, Region, State) creates a risk of delays in project approvals. Since many grants (including Conto Termico) require works to be completed within strict timeframes (e.g., 12–18 months after booking), any bureaucratic delay could result in the loss of funding. Heritage protection risks are particularly acute: unpredictable requirements from heritage committees could increase the CAPEX of the historical cluster by 20–30% due to the need for more expensive materials.

4.4.2. Cost Inflation and Resource Availability

The Italian construction sector in 2024–2025 remains under pressure from a shortage of skilled labor and volatility in material prices (insulation, electronics). An expected 6.7% increase in labor costs in the coming years

could significantly undermine initial budgets. Additionally, long lead times for grid connection of new PV systems may delay the onset of revenue from energy generation.

4.4.3 Cash Flows and Asset Value

The investment logic of PED UNIPD is based on the "Life Cycle Cost" concept. While the initial investment (CAPEX) is very high, it protects the university from future price shocks in the energy market. Research indicates that buildings undergoing deep renovation command a market premium of 13.5% compared to non-modernized properties. For the university, this translates into an increase in the capitalization of its real estate portfolio, which improves the institution's credit rating.

5. Strategic Integration and Phasing

The transition of the UNIPD campus into a Positive Energy District (PED) requires a highly coordinated approach that balances technical ambitions with strict heritage laws, financial realities, and the social dynamics of 75,000 campus users. This section integrates the engineering, social, institutional, and financial dimensions into a unified, phased masterplan.

5.1. Identification of Priority Measures

Before deploying capital-intensive engineering solutions, foundational institutional and financial frameworks must be established. The immediate priority measures include:

- Securing Capital: Initiating applications for the Conto Termico 3.0 grant program and engaging Energy Service Companies (ESCOs) to fund the €134.2 million total CAPEX without over-leveraging the university's balance sheet.
- Institutional Alignment: Establishing joint coordination forums with the City of Padova and regional heritage authorities to secure pre-approvals for retrofits and align with the "Net Zero Cities" 2030 municipal target.
- Behavioral Quick-Wins: Launching the "Green Ambassadors" student gamification program to immediately address the 30–50% energy performance gap caused by occupant behavior, requiring minimal initial capital.

5.2. Short-, Medium-, and Long-Term Implementation Phases

To mitigate regulatory risks and accommodate multi-year budgeting cycles, the physical and social interventions are structured into three chronological phases:

- Short-Term Phase (0–1 Years): Foundation & Awareness The focus is strictly on securing financing, baseline auditing, and changing user behavior. Non-invasive operational upgrades will be applied, such as standardizing seasonal HVAC changeovers and introducing "Comfort vs. Safety" co-design workshops with academic lab researchers.
- Medium-Term Phase (1–3 Years): Conservative Modernization Interventions will target the Historic Cluster. Capitalizing on an estimated €27.5 million allocation, internal hydraulic networks will be replaced, high-efficiency heat pumps installed, and Building Automation and Control Systems (BACS) deployed. This yields a rapid ROI through automated energy management without triggering prolonged heritage legal battles.
- Long-Term Phase (3–10 Years): Deep Renovation & Renewable Integration The most capital-intensive phase (€103 million) targets the Modern Cluster. It involves full envelope insulation and window replacements to achieve a 38% reduction in heating demand. Simultaneously, the 2,883 kWp Solar PV

system will be deployed across the newly modernized roofs to maximize the district's renewable generation capability.

5.3. Explicit Discussion of Trade-Offs Between Dimensions

Achieving PED status within a dense, historical urban environment necessitates strategic compromises across the four dimensions:

- **Engineering vs. Institutional (Heritage):** A massive trade-off exists between maximizing energy efficiency and preserving cultural identity. European mandates demand near-zero energy performance, yet strict Italian heritage laws legally prohibit external insulation on historic facades. The strategy compromises by accepting a lower energy savings yield (15%) in the Historic Cluster, forcing a heavier reliance on mechanical heat pumps rather than passive envelope physics.
- **Financial vs. Technical Timeline:** The technical potential exists to renovate the entire Modern Cluster immediately; however, the financial dimension dictates a slower reality. Because the €103 million deep renovation relies heavily on multi-year Conto Termico disbursements, massive technical interventions must be deliberately delayed to the Long-Term Phase to match public funding liquidity constraints.
- **Social vs. Engineering Efficiency:** Imposing strict, automated limits on laboratory ventilation and dorm heating would guarantee technical energy reductions. However, this trades away social equity and scientific safety. The strategy prioritizes user comfort and research integrity by opting for collaborative co-design over top-down technological enforcement, accepting that behavioral change is slower but more sustainable.

5.4. Clear Justification Toward Energy Positivity

The proposed pathway systematically drives the district toward positivity by sequentially attacking both demand and supply. The combined deep renovations and conservative upgrades successfully compress the district's total annual energy demand from a baseline of 26,893 MWh down to 24,320 MWh. Following this reduction, the 14,415 m² of modern rooftop space is saturated with solar PV, injecting 3,942 MWh/year of clean electricity directly into the local grid.

While the on-site physics cap the Renewable Energy Share (RES) at 16.2% due to the campus's constant 24/7 electrical baseload and limited historic roof space, the institutional pathway provides the final bridge to true positivity. By officially participating in Renewable and Solidarity Energy Communities (RSEC) alongside the City of Padova, UNIPD will effectively expand its PED boundary. This regulatory innovation allows the university to import the remaining 20,377 MWh from external municipal solar arrays, officially closing the energy deficit and cementing its role as a socially, financially, and technically viable Positive Energy District.

Bibliography

[1] University of Padua (UNIPD), "Sustainability Report 2023-2024," University of Padua Sustainable Development, Padua, Italy, 2024. [Online]. Available: https://www.sostenibile.unipd.it/wp-content/uploads/2025/01/Web-Report-2024-en.html

[2] Institut Wohnen und Umwelt (IWU), "TABULA WebTool: Typology Approach for Building Stock Energy Assessment," EPISCOPE Project, Intelligent Energy Europe. [Online]. Available: https://webtool.building-typology.eu/.

[3] National Renewable Energy Laboratory (NREL), "U.S. Solar Photovoltaic System and Energy Storage Cost Benchmark," Dept. of Energy, Golden, CO, USA, Tech. Rep. NREL/TP-6A20-80694, 2022. [Online]. Available: https://www.nrel.gov/docs/fy22osti/80694.pdf

[4] European Commission Joint Research Centre (JRC), "Photovoltaic Geographical Information System (PVGIS)," 2023. [Online]. Available: https://re.jrc.ec.europa.eu/pvg_tools/en/.

[5] Italian Organization for Standardization (UNI), "UNI 10349-1:2016: Heating and cooling of buildings - Climatic data," Milan, Italy, 2016. [Online]. Available: https://store.uni.com/uni-10349-1-2016

[6] Italian Institute for Environmental Protection and Research (ISPRA), "Atmospheric emission factors of greenhouse gases and other pollutants from power sector," Rome, Italy, ISPRA Rep. 280/2018. [Online]. Available:https://www.isprambiente.gov.it/en/publications/reports/atmospheric-emission-factors-of-greenhouse-gases-and-other-pollutants-from-power-sector

[7] ESU of Padua. Available: https://www.esu.pd.it/en/residences/

[8] Habtamu Tkubet Ebuy, H. B. (2023). Occupant Behavior Impact on Building Sustainability Performance: A Literature Review. Vandœuvre-Lès-Nancy: Sustainability. Available: https://www.mdpi.com/2071-1050/15/3/2440

[9] Hera Group. Available: https://eng.gruppohera.it/group_eng/who-we-are/the-hera-group-in-the-territory

[10] Kivimaa, P. (2025). Opportunities and challenges for EU energy policy 2025-2035 – a systemic perspective. Seville: Joint Research Centre. Available: https://publications.jrc.ec.europa.eu/repository/handle/JRC142668

[11] University of Padua. Available: https://www.unipd.it/en/university-padua